AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Biology Paper I

Time allowed: 35 minutes Marks 25

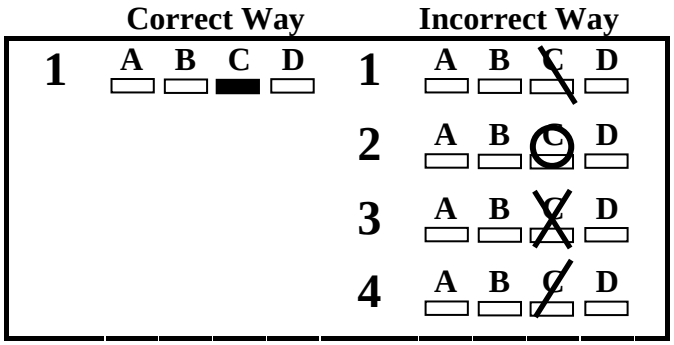
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

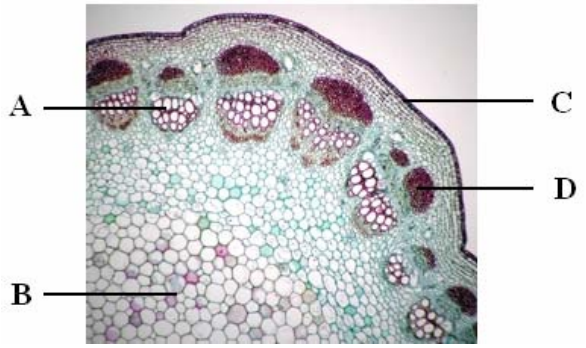
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes

1. Identify phloem tissues in the given diagram of a dicotyledonous stem.



2. Which of the following is correct about the structure and function of xylem vessels?

	Structure	Function
A.	Hollow tubes	Transport of water
B.	Hollow tubes	Transport of sugar
C.	Discontinuous lumen	Transport of water
D.	Discontinuous lumen	Transport of sugar

3. All of the following are characteristics of red blood cells EXCEPT

A. they are produced in liver.
B. they contain haemoglobin.
C. they do not have a nucleus.
D. they transport oxygen.

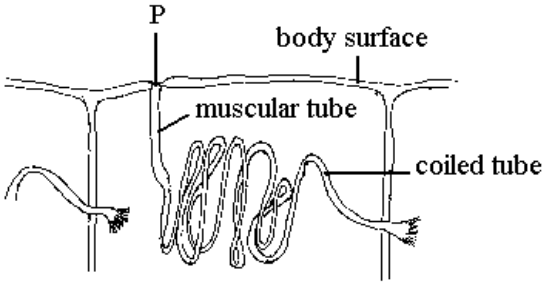
4. The daytime release of oxygen by plants is considered as a waste because it is produced as a result of

A. exchange of gases.
B. synthesis of glucose.
C. break down of starch.
D. removal of excess water.

5. The diagram shows the nephridium of an earthworm.

Which is the correct identification of the structure labelled “P” and its function?

	Identification	Function
A.	Nephrostome	Removal of waste
B.	Nephridiopore	Removal of waste
C.	Nephridiopore	Ultra filtration
D.	Nephrostome	Ultra filtration



6. The amount of glucose in the renal artery is more than in the renal vein due to the

   A. excretion of glucose in the urine.
   B. utilization of glucose by kidney cells.
   C. storage of glucose in the renal pelvis.
   D. filtration of glucose in the glomerulus.

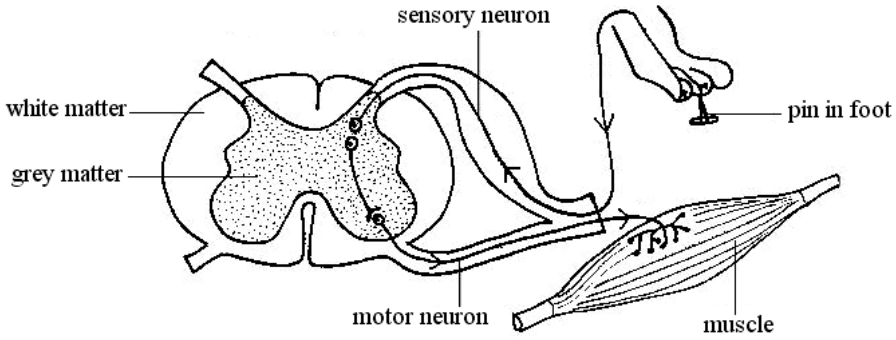
7. Which of the following tissues provides flexible support to plants?

   A. Tracheids
   B. Parenchyma
   C. Collenchyma
   D. Vascular cambium

8. What is the function of tendons in the body?

   A. Join one bone to another
   B. Attach muscles to the bones
   C. Reduce friction between the bones
   D. Conduct nerve impulses to the bones

9. The diagram shows a portion of a spinal cord.



   The diagram has an error in the

   A. position of motor neuron.
   B. structure of sensory neuron.
   C. direction of the nerve impulses.
   D. labelling of the regions of spinal cord.

10. Electric wires are coated with rubber for insulation. Which part of a neuron serves the same function?

   A. Dendrites
   B. Neurilemma
   C. Myelin sheath
   D. Nodes of Ranvier

11. Which of the following parts of an eye are involved in focus on a near object?

    A. Ciliary muscles, pupil and lens
    B. Ciliary muscles, cornea and lens
    C. Suspensory ligaments, pupil and cornea
    D. Ciliary muscles, suspensory ligaments and lens

12. A long, thin stem which creeps along the surface of the soil is called a

    A. runner.
    B. sucker.
    C. tendril.
    D. stock.

13. A plant with large and brightly coloured petals is most likely to be pollinated by

    A. insects.
    B. water.
    C. wind.
    D. soil.

14. Double fertilization in plants results in the formation of

    A. one haploid zygote.
    B. two haploid zygotes.
    C. one triploid endosperm cell.
    D. two triploid endosperm cells.

15. The process of crossing over during meiosis is significant because it

    A. supplies exact DNA sequence of the parents.
    B. provides genetic variation in offspring.
    C. allows the formation of diploid cells.
    D. reduces the time for cell division.

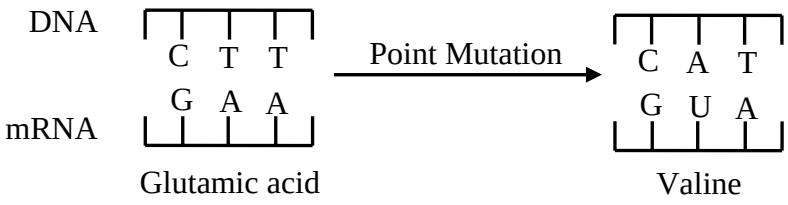
16. For the trait of height, if tallness is represented by “T” then its partner gene “t” will be called

    A. allele.
    B. dominant.
    C. phenotype.
    D. homozygous.

17. A homozygous round seed plant is crossed with a wrinkled seed plant. What is the percentage of wrinkled seed plants appearing in F1 generation?

    A. 100%
    B. 50%
    C. 25%
    D. 0%

18. Which disease is caused by the following mutation?

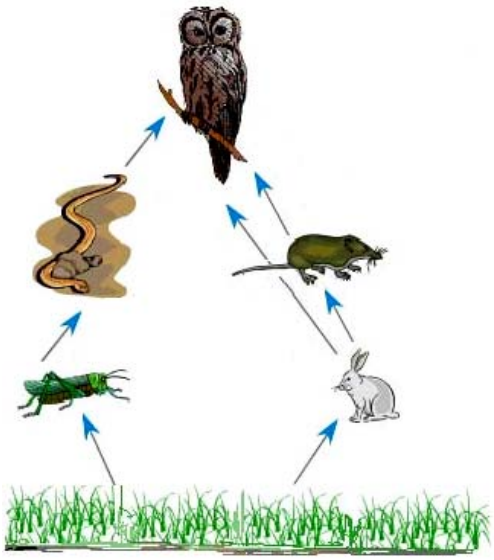


A. Haemophilia
B. Colour blindness
C. Down's syndrome
D. Sickle cell anaemia

19. The sepals, petals, stamens and carpel are considered to have evolved from leaves. In this way, the parts of a flower are an example of

A. homologous organs.
B. convergent organs.
C. analogous organs.
D. vestigial organs.

20. The diagram shows a food web.



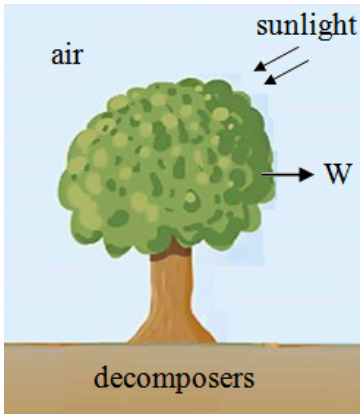
Which organism receives the least energy in the given food web?

A. Grass hopper
B. Snake
C. Owl
D. Rat

21. Why is oxygen considered as a renewable resource?

A. It is given off by fungi during decomposition.
B. It is an end product of metabolism in plants.
C. It is recycled by bacteria in the air.
D. It is refilled in the air by sunlight.

22. The diagram shows parts of a carbon cycle.



Which of the following carbon containing compounds is found at point W?

A. Urea
B. Starch
C. Methane
D. Carbon dioxide

23. Barnacles (marine crustaceans) attach themselves to the backs of whales and get a ride to better feeding places. Such ecological interdependence is an example of

A. mutualism.
B. predation.
C. parasitism.
D. commensalism.

24. The table shows some diseases with their respective causative agents.
For which disease will treatment through antibiotics be most appropriate?

	Disease	**Causative agent**
A	Influenza	Virus
B	Ring worm	Fungus
C	Jacob's disease	Prion
D	Typhoid	Bacteria

25. Biological knowledge has enabled farmers to overcome many problems associated with sericulture.
Which of the following practices makes the process easy and productive?

A. Produce wingless variety of silkworms.
B. Smaller number of silk worms should be used.
C. Introduce some artificial silk into the medium.
D. Add insecticide to prevent microbial contamination.

END OF PAPER

Please use this page for rough work